

BUCURESTI FILARET, Romania

WMO: 154220

Lat: 44.4122N

Lon: 26.0939E

Elev: 274

StdP: 14.55

Time Zone: 2.00 (EUE)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.2	18.6	9.6	9.1	17.3	13.3	10.9	20.7	27.9	31.4	26.0	31.2	3.4	220	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.9	94.6	71.2	92.0	70.0	89.1	69.1	74.4	88.4	72.7	86.3	71.3	84.6	2.5	40

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.6	113.9	78.5	68.7	106.4	77.1	67.2	101.0	76.1	38.0	87.9	36.5	86.3	35.2	84.9	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.2	13.6	9.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.2	30.7	38.9	46.5	54.4	64.4	73.2	75.5	76.5	67.3	55.1	43.1	35.9
	DBStd	16.97	7.85	8.03	7.93	7.71	5.57	5.29	4.47	5.32	6.63	7.70	7.95	6.47
	HDD50	1834	598	319	158	45	0	0	0	0	0	41	235	438
	HDD65	4699	1063	731	573	322	77	5	0	2	54	313	657	902
	CDD50	3739	0	8	50	178	448	695	792	822	519	199	28	0
	CDD65	1130	0	0	0	5	59	250	327	359	124	6	0	0
	CDH74	10758	0	1	7	107	570	2253	3031	3629	1072	87	1	0
	CDH80	4509	0	0	0	16	123	891	1308	1778	379	14	0	0
Wind	WSAvg	3.2	3.8	3.9	4.2	3.5	3.1	2.6	2.4	2.4	2.5	2.8	3.1	3.8
Precipitation	PrecAvg	24.1	1.7	1.6	1.5	1.8	2.8	3.1	2.2	2.3	1.7	1.3	2.1	1.9
	PrecMax	33.6	5.2	4.3	5.0	3.8	10.2	7.0	5.3	6.9	6.1	5.6	8.1	7.3
	PrecMin	16.2	0.2	0.3	0.1	0.2	0.4	0.6	0.3	0.1	0.0	0.0	0.0	0.0
	PrecStd	4.3	1.3	1.1	1.3	0.9	2.0	1.7	1.3	1.6	1.6	1.1	1.7	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	52.8	66.2	74.5	82.3	87.1	96.4	96.4	98.6	92.4	81.5	70.5	56.1
		MCWB	46.4	54.5	57.1	63.2	66.7	73.7	72.5	71.3	68.2	64.0	58.5	48.4
	2%	DB	47.4	58.6	68.5	77.0	83.4	92.1	93.5	95.2	88.3	75.6	62.9	51.6
		MCWB	43.1	50.2	53.7	61.1	66.0	72.6	70.8	70.9	66.8	61.1	55.3	46.7
	5%	DB	43.3	54.0	64.1	72.0	80.4	88.2	90.7	92.6	83.8	71.2	57.7	47.5
		MCWB	40.6	47.4	51.5	57.9	64.6	71.0	70.0	69.5	65.7	58.9	52.4	43.9
	10%	DB	40.3	49.8	59.6	68.5	76.8	85.1	87.2	89.8	79.9	67.2	53.6	44.0
		MCWB	38.2	44.9	49.4	56.3	62.3	69.5	68.8	68.9	64.4	57.3	50.8	41.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	55.5	58.0	64.5	69.6	77.3	75.6	76.5	71.8	65.1	59.3	50.4
		MCDB	52.2	64.9	71.2	79.7	81.7	91.9	88.7	90.0	86.4	76.4	68.2	54.0
	2%	WB	43.6	51.5	54.7	62.1	67.8	74.5	73.6	73.8	68.9	62.5	56.5	47.0
		MCDB	46.6	58.5	66.1	75.3	79.7	87.9	87.4	88.6	81.6	71.4	62.7	51.1
	5%	WB	41.1	48.0	52.4	59.8	65.9	72.5	72.1	72.0	67.1	60.7	53.9	44.4
		MCDB	43.3	52.6	62.3	70.5	77.5	85.1	85.2	86.9	79.8	69.0	57.4	47.2
	10%	WB	38.5	45.4	50.0	57.3	64.1	70.9	70.8	70.6	65.5	58.4	51.3	42.1
		MCDB	40.2	48.9	58.2	65.9	74.6	82.7	83.2	85.5	77.8	65.3	53.7	44.3
Mean Daily Temperature Range	5% DB	MDBR	11.1	14.4	17.5	19.5	20.6	20.2	21.0	22.9	21.0	18.2	11.7	10.3
		MCDBR	15.1	21.6	26.1	27.0	25.6	24.4	25.7	26.7	26.6	25.6	19.5	16.3
		MCWBR	10.3	13.5	13.3	13.6	11.0	9.7	8.8	8.7	9.8	12.6	11.7	10.9
	5% WB	MCDBR	13.6	19.1	24.0	25.6	23.3	23.0	22.2	24.1	23.2	23.3	16.8	17.0
	MCWBR	9.8	12.2	13.0	13.2	10.6	10.8	8.8	8.8	9.3	12.1	10.5	11.2	
Clear-Sky Solar Irradiance	taub	0.349	0.381	0.416	0.451	0.426	0.436	0.446	0.460	0.424	0.403	0.376	0.352	
	taud	2.268	2.186	2.146	2.078	2.226	2.233	2.213	2.167	2.233	2.261	2.286	2.281	
	Ebn at Noon	236	249	256	257	269	266	261	252	251	239	224	221	
	Edh at Noon	28	36	42	49	43	43	44	44	38	33	27	25	
All-Sky Solar Radiation	RadAvg	439	703	1088	1492	1862	2014	2041	1821	1321	857	472	362	
	RadStd	68	106	116	154	138	140	134	100	100	103	62	55	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (62 neighbors)	+0.83	N/A	N/A	N/A	N/A	N/A	N/A	-221	+126	+72	

Nomenclature: See separate page